2026-27 DIGITAL EDITION

KNEE PAIN & ARTHRITIS

The Indian Playbook

Read your own X-ray. Build the strength that protects the joint.
Know which injections, supplements and operations are worth your money.
Verified against ICMR, NICE, OARSI, NPPA and the 2025 Indian evidence.

20 CHAPTERS | 28 TREATMENTS AUDITED | 7 PRINTABLE WORKSHEETS

The eight-week strength programme - the real cost of a knee replacement
Steroid, PRP and stem cell injections - the Indian home, rebuilt for your knees
Flare plans - red flags - doctor visit sheets - surgery decision checklist

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How to use this book

Knee pain is the most common reason Indians over 45 visit an orthopaedic clinic, and it is also the condition with the widest gap between what the evidence says and what patients are sold. This book closes that gap in about three hours of reading.

Read Part I to understand what is happening and why it is not simply wear and tear. Read Part II before you agree to any scan. Read Part III, because it contains the eight-week programme that does most of the work, and the weight arithmetic that does the rest. Parts IV and V are where the money is, and where the largest avoidable spending in Indian knee care sits. Part VI is for people whose knees do not yet hurt much, and for the bad days.

The worksheets at the back are designed to be printed. If you take one thing to your next appointment, take Worksheet 1 and Worksheet 4.

What this book is and is not

It is an educational guide, written for Indian readers, drawing on the clinical guidelines that Indian and international doctors actually work from, and on Indian community studies, government price notifications and company disclosures where those are the only source. It is not a prescription, a diagnosis, or a substitute for a clinician who can examine your knee. Nothing here overrides the advice of the doctor treating you, and nothing here is a reason to stop a medicine you have been prescribed.

Prices quoted are indicative ranges at the time of writing and change constantly. Guideline positions are current as of the 2024 and 2025 editions cited in the sources. Drug doses are deliberately not prescribed here, because the right dose depends on your kidneys, your heart, your other medicines and your weight, and that is a conversation with a doctor, not a page in a book.

If you have diabetes, kidney disease, heart failure, a bleeding disorder, or you are pregnant, several of the medicines discussed here need a specific conversation with your doctor before you use them. The book flags where. It cannot make that decision for you.

A note on sources

Every statistic in this book is traceable to a document listed in the final chapter. Where a number comes from a company statement or a press report rather than a peer-reviewed guideline, the text says so plainly. Where two credible sources disagree, both are given, because a book that pretends to certainty about a contested question is not worth your money.

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The one-page version

If you read nothing else in this book, read this page. Every claim below is unpacked, sourced and costed in the chapters that follow.

1. **Your knee pain is probably not "wear and tear" from old age.** A pooled analysis of 16 Indian community studies puts knee osteoarthritis at 20.24% of adults; in people over 60 it is 47%. It is not an old person's disease, and it is not rare.
2. **You do not need an MRI to be diagnosed.** NICE, the UK's guideline body, says osteoarthritis is diagnosed in the clinic: age 45 or over, pain that comes with activity, and morning stiffness lasting 30 minutes or less. Imaging is for atypical cases.
3. **The one scan that matters is a standing X-ray.** Weight-bearing AP plus a lateral view, both knees. A lying-down X-ray under-reads the joint space and has produced thousands of unnecessary MRI scans.
4. **Exercise is the treatment, not the thing to avoid.** The ICMR's own Standard Treatment Workflow for knee osteoarthritis lists quadriceps strengthening and hamstring stretching first, before any medicine. NICE offers therapeutic exercise to everyone with the condition.
5. **Every kilogram you lose takes about four kilograms off each step.** A 10% weight loss beats 5%, and 5% beats nothing. This is the single largest lever you control.
6. **Topical gel before tablets.** Both NICE and OARSI recommend a topical NSAID for knee osteoarthritis. Oral painkillers are for short courses, at the lowest dose, and opioids are actively not recommended.
7. **Three of the most sold products for knees do not work.** Glucosamine, chondroitin and collagen have no guideline support. The American College of Rheumatology strongly recommends against glucosamine.
8. **An injection is a pause button, not a repair.** Steroid injections work for weeks, not years, and the accepted ceiling is three to four in the same knee per year. PRP has no supportive guideline and is under regulatory scrutiny in India.
9. **Surgery is for a specific picture, not a grade on a report.** The ICMR lists the indications: Kellgren-Lawrence grade 4 plus pain that limits daily life, after conservative treatment has failed. The same document is headed "keep a high threshold for invasive procedures".
10. **The implant inside you is price-capped by law.** NPPA caps a primary knee replacement system at Rs 51,563, extended to 15 November 2026. Ask for the implant sticker and the invoice line.

> **What this book will not do**
>
> It will not diagnose you, prescribe for you, or replace an examination by a clinician who can feel your knee. It will do something narrower and more useful: put the same evidence your doctor is working from into your hands, in plain language, with Indian prices and Indian daily life attached. You will leave consultations with better questions, and you will stop paying for things that do not work.

1. Is it arthritis, or something else?

Most knee pain in India is not arthritis. Of the people who arrive at an orthopaedic OPD convinced they have "ghutna ghis gaya", a meaningful share have a treatable soft-tissue problem that will resolve in weeks if it is named correctly.

The word arthritis gets used for everything from a strained ligament to advanced joint failure. That matters, because the treatment for each is different and sometimes opposite. Rest is right for an inflamed bursa and wrong for osteoarthritis. Strengthening is right for osteoarthritis and wrong for an acute ligament tear. Getting the label right is the whole game.

Start with what your knee actually does, not what it looks like on a report. The table below maps the patterns clinicians use in a ten-minute examination. Find the row that sounds most like you, then read the chapter that follows. This is not a diagnosis. It is a way of knowing what to ask.

Eight patterns that explain most knee pain

Pattern	What it feels like	What usually gives it away	Usual first move
Knee osteoarthritis	Ache on the inner side of the knee, worse after walking, squatting or climbing stairs. Stiff for a few minutes after sitting.	Age 45-plus, morning stiffness under 30 minutes, crackling on movement, bony enlargement at the joint line.	Strengthening programme plus weight work. See Chapters 6 and 7.
Patellofemoral pain	Pain at the front, around or under the kneecap. Worse going down stairs, sitting with knees bent, or standing after a long drive.	Pain reproduced by pressing the kneecap or by a squat; often a young or middle-aged desk worker.	Hip and quadriceps strengthening, load management. Not surgery.
Meniscus tear	Sharp pain on one side, catching or locking, a feeling the knee will give way on twisting.	A specific twist or deep squat, swelling within 24 hours, pain on the joint line.	Examination first. MRI only if locking or true giving-way. See Chapter 4.
IT band syndrome	Sharp pain on the outer side, often in runners and cyclists, worse on downhill runs.	Tender point about 2-3 cm above the outer joint line; no swelling.	Load reduction, gluteal strengthening, gait review.
Pes anserine bursitis	Ache on the inner side, a few centimetres below the joint line, worse at night lying on that side.	Local tenderness in one spot; common with obesity, diabetes and knee arthritis.	Ice, topical gel, short course of oral anti-inflammatory, treat the underlying knee.
Patellar tendinitis	Pain just below the kneecap, worse after jumping or stairs, stiff first thing in the morning.	Tender tendon, pain on resisted knee extension.	Progressive loading of the tendon. Rest alone makes it worse.
Referred hip pain	Pain felt at the knee or thigh, often with a limp, sometimes with groin pain.	Reduced hip rotation on examination. Common in older adults with hip arthritis.	Hip examination and hip X-ray. Treating the knee will fail.
Inflammatory arthritis	Both knees, swelling, morning stiffness lasting more than an hour, fatigue, sometimes fever.	Symmetrical swelling, other joints involved, young to middle-aged women most often.	Rheumatology referral and blood tests within weeks. Do not wait months.

Two more patterns are emergencies rather than diagnoses. A knee that becomes hot, red, very swollen and painful over hours, with fever and an inability to put weight on it, may be infected. That is a same-day hospital problem, not a wait-and-watch problem. A calf that swells and hurts after surgery, a long flight or a period of immobility may be a clot, which is also same-day.

The sixty-second self-check

Answer these five before your appointment. Write the answers down; memory drifts and consultations are short.

- **When is it worst?** First steps in the morning, after a long walk, on stairs, at night in bed, or all the time? Night pain and rest pain are different signals from activity pain.
- **How long is it stiff after rest?** Under 30 minutes points to osteoarthritis. Over an hour, especially with swelling, points towards an inflammatory arthritis and changes who you should see.
- **Has it swollen?** Swelling within 24 hours of an injury suggests bleeding or a meniscal tear. Gradual swelling over days is more typical of osteoarthritis or bursitis.
- **Does it lock or give way?** True locking, where the knee physically will not straighten, is a mechanical problem. Giving way can be weakness, a ligament, or simply pain inhibition.
- **What else is going on?** Diabetes, thyroid disease, recent weight gain, menopause, a fall, a new gym programme, a change of footwear, a long drive, a wedding with three days of floor seating.

> **Why the label matters more than the scan**
>
> A 52-year-old woman with pain on the inner knee, fifteen minutes of morning stiffness and no locking has osteoarthritis until proven otherwise. She needs a strength programme, a weight conversation and a topical gel. Send her for an MRI and she will be told she has a meniscal tear and a cartilage defect, because almost every 52-year-old knee has both on MRI. The scan has not added information. It has added a decision that did not need to be made.

2. Indian knees wear out earlier, and the data says why

India does not have a joint registry or a national arthritis database. What it has is a set of community surveys that, put together, describe one of the largest orthopaedic problems in the country.

20.24%
Pooled knee OA prevalence
16 Indian community studies, meta-analysis 2025

47%
Prevalence over age 60
14 studies, 5,029 participants, 2025

48.4m
Indians with knee OA
Cipla company statement, 2021 estimate

250,000+
Knee replacements a year
Reported by Business Standard, 2025

The pooled figure of 20.24% comes from a 2025 systematic review of sixteen population-based Indian studies, with a wide confidence interval of 11.90% to 30.13% because the studies used different definitions and settings. The authors found prevalence slightly higher in rural (21.25%) and mixed (22.38%) populations than urban (15.42%), though the difference was not statistically significant. In people aged 60 and above, a separate 2025 meta-analysis of fourteen studies and 5,029 participants found 47%, with women at 50.7% and men at 26.2%.

A five-site study across big cities, small cities, towns and villages, covering 5,000 people with X-ray confirmation, found 28.7% overall and 31.6% in women, with obesity, age and sedentary work each significantly associated. A rural Tiruvallur study of 427 adults aged 40 and above found 34.6%. These numbers do not agree with each other, and that is the honest picture: somewhere between one in five and one in three Indian adults has radiographic knee osteoarthritis.

What the Indian risk factors actually are

The Tiruvallur study is worth reading closely because it measured the things that are specific to how Indians live. The odds ratios below are associations, not proof of cause, but the pattern is hard to argue with.

Factor	Odds ratio	What it means for you
Attained menopause	21.54	The largest single association in the study. If you are a woman in your late forties or fifties with new knee pain, this is the frame, not a personal failure.
Diabetes	16.25	Chronic high glucose affects cartilage and tendon. If your HbA1c is high, knee pain and blood sugar are one problem, not two.
Indian-style toilet	12.26	The deepest knee bend of the average Indian day, performed several times daily, often with no support to hold.
Obesity	5.29	Every kilogram is roughly four kilograms of load per step. See Chapter 9.
Hypertension	5.01	Partly shared lifestyle, partly vascular. Also a reason to be careful with oral anti-inflammatories.
Sedentary lifestyle	3.26	Sitting is not rest for a knee. It deconditions the muscle that protects it.
Prolonged sitting	2.51	Long desk days, long drives, long shifts.
Tobacco use	2.27	Slower tissue healing and worse outcomes after surgery.

Two of these deserve a second look. The Indian toilet finding is not an argument for Western bathrooms. It is an argument for a Rs 1,000 commode stool, a grab bar, and a household that stops treating a deep squat as the only way to use a bathroom. Chapter 10 shows the swaps.

The menopause finding explains a pattern every orthopaedic surgeon in India sees: women in their forties arriving with knees that hurt far more than their X-rays suggest. Oestrogen withdrawal affects cartilage, muscle mass and bone. It does not mean you are destined for surgery. It means the strengthening work in Chapter 7 matters more for you, earlier.

Why younger Indians are now showing up with it

Several Indian newspapers reported through 2025 and 2026 that orthopaedic clinics are seeing more knee pain in people under 40, with surgeons attributing part of the rise to post-Covid weight gain, longer sitting hours and reduced activity. That is clinician impression reported in the press, not a population survey, so treat it as a signal rather than a statistic. Cipla's own launch material for its knee therapy noted rising prevalence "even among younger populations".

What is not in doubt is the mechanism. A knee tolerates load it is trained for. It does not tolerate a 12-kilogram weight gain over three years combined with a job that removed 8,000 steps a day from your life. The joint did not fail. The load-to-capacity ratio did. Chapter 18 is the prevention chapter for anyone under 40 who wants to keep the knee they have.

> **The number to remember**
>
> One in five Indian adults has knee osteoarthritis on X-ray, and half of Indians over 60 do. If your knee hurts, you are not unlucky and you are not alone. You are in the largest single group of musculoskeletal patients in the country, which is exactly why the good information is worth having.

3. What is actually happening inside your knee

The phrase "cartilage has worn away" has done more damage to Indian knees than any single piece of medical misinformation, because it implies a mechanical problem with one fix: stop using the joint.

Cartilage has no nerve supply. It cannot hurt. The pain in an arthritic knee comes from the structures around it: the synovial lining that becomes inflamed, the joint capsule, the bone underneath the cartilage, the meniscus, the ligaments and the muscles inserting around the joint. This is why two people with identical X-rays can have completely different pain, and why the person with the worse-looking X-ray is not always the one who suffers most.

It also explains why rest is the wrong prescription. Muscle is the shock absorber that decides how much force reaches those pain-sensitive structures. Take six weeks off your feet and you lose quadriceps strength; lose quadriceps strength and the same walk delivers more force to a joint that is already complaining. The pain gets worse, so you rest more. That loop is the actual disease process for most people, and it is breakable without surgery.

Kellgren-Lawrence grades, in plain language

Radiologists grade knee osteoarthritis on a five-point scale from 1957 that is still used everywhere, and the ICMR's Standard Treatment Workflow reproduces it. It describes what is visible on an X-ray. It does not describe your pain.

Grade	What the report says	What it means	What it does not mean
0	No radiographic features of OA	A normal-looking knee for your age.	That your pain is imaginary. Soft tissue problems do not show on X-ray.
1	Doubtful joint space narrowing, possible bony spurring	Very early change. Often nothing to treat beyond exercise.	That you are on a countdown to surgery.
2	Definite bony spurs, possible narrowing	Mild osteoarthritis. This is where a strength programme changes the next decade.	That you need an injection or an MRI.
3	Multiple spurs, definite narrowing, denser bone, possible deformity	Moderate osteoarthritis. Symptoms usually track this grade more closely.	That grade 3 automatically means replacement.
4	Large spurs, marked narrowing, severe sclerosis, definite deformity	Severe osteoarthritis on imaging.	That you need surgery today. The ICMR requires failed conservative treatment and pain that limits daily life as well.

NICE is explicit that management should be guided by symptoms and physical function, not by the image. Read that again, because it is the sentence that will save you the most money. A grade 4 knee that lets you walk to the market without much pain does not need an operation. A grade 2 knee that stops you climbing your own stairs deserves aggressive treatment.

Three things that are true about arthritic knees

- **It is not a bearing wearing out.** Osteoarthritis involves the whole joint: low-grade inflammation of the synovial lining, changes in the bone beneath the cartilage, weakening of the muscles around it. Calling it wear and tear leads to the wrong treatment.
- **It is not always progressive.** Many people stabilise or improve, particularly with strength work and weight loss. The idea that it only ever gets worse is not supported by long-term studies.
- **Pain intensity is not damage intensity.** Flares are driven by inflammation, load and muscle fatigue. That is why a bad week does not mean your knee deteriorated, and why the flare plan in Chapter 19 works.

> **What the crackling sound is**
>
> Crepitus, the crackling or grating under your hand, is gas and rough surfaces moving past each other. It is common in knees with no pain at all and it is not a sign that you are "grinding the joint away". If it comes with pain and swelling, it is worth mentioning. On its own, it is noise.

4. The expensive diagnosis you probably do not need

NICE does not recommend routine imaging to diagnose osteoarthritis. It recommends a clinical diagnosis, made in the consultation room, in anyone 45 or over with activity-related joint pain and morning stiffness lasting 30 minutes or less.

That single recommendation, if applied in India, would remove a very large share of the knee MRIs currently being done. Not because scans are bad, but because in a typical osteoarthritis presentation the scan does not change what should happen next. You will still be offered exercise, weight management and a topical gel. You will simply have paid for a picture of it.

What a proper ten-minute examination contains

You can tell within two minutes whether the person examining you is looking for a diagnosis or confirming one. A careful examination includes all of the following. If three or four are missing, ask for them.

- Watching you walk, and watching you stand with both feet visible, to see alignment and any varus (bow-legged) drift.
- Feeling for fluid in the joint, and measuring how far the knee bends and straightens with a goniometer or by eye.
- Pressing along the joint line on both sides, and pressing the kneecap to check the patellofemoral joint.
- Testing the menisci and the collateral and cruciate ligaments if there was an injury.
- Looking at the quadriceps for wasting, and testing whether you can do a straight leg raise.
- Examining the hip, because hip arthritis refers pain to the knee and is missed constantly.
- Checking the other knee, the hands and the feet if inflammatory arthritis is possible.
- Asking about night pain, rest pain, fever and weight loss.

The hip examination is the one to insist on. An older adult with knee pain and a normal knee X-ray very often has hip osteoarthritis, and a knee replacement performed on that patient fails to help because the pain was never coming from the knee.

The only X-ray views worth paying for

The ICMR workflow specifies bilateral knee X-rays, anteroposterior standing views plus lateral views. The word standing is doing the work. A weight-bearing AP view shows the joint space as it actually behaves when you are on your feet. A supine view taken lying on a table can look normal in a knee that has significant narrowing under load.

Test	When it is genuinely useful	When it is wasted money
Standing AP and lateral X-ray, both knees	Almost every first presentation of knee pain over 40. Cheap, fast, shows alignment and grade.	Rarely wasted, if taken standing.
MRI without contrast	Locking or true giving-way suggesting a meniscal tear; suspected ligament injury; persistent pain with a normal X-ray in a younger person; suspected infection or tumour; pre-surgical planning.	Routine "how bad is my arthritis" requests. It will show cartilage and meniscal changes in almost everyone over 50, and those findings are often unrelated to the pain.
Blood tests: ESR, CRP, RA factor, anti-CCP, uric acid	Symmetrical swelling, stiffness over an hour, fever, young women, suspected gout.	Routine ordering for typical mechanical knee pain.
HbA1c, vitamin D, thyroid	When diabetes, deficiency or thyroid disease is plausible, which in India is often.	As a substitute for examining the knee.
Diagnostic ultrasound	Guiding an injection, assessing a bursa or a tendon, checking for effusion.	As a general arthritis test.

Before agreeing to any scan, ask one question: what will change depending on the result? If the answer is "we will do the same thing but we will know", you are buying reassurance, which is legitimate but should be a conscious purchase rather than a default one. If the answer is "if there is a tear we will operate, and if there is not we will do

physiotherapy", the scan is doing real work.

> **The three-question scan test**
> One: what are we looking for? Two: if you find it, what will we do differently? Three: if we find nothing, what will we do? A scan that cannot change the plan is a photograph, not a test.

5. Reading your own X-ray report without panicking

Radiology reports are written for referring doctors, in a vocabulary designed to be precise rather than kind. Here is what the common phrases actually mean, and what to do about each.

What the report says	What it means	What to do
"Marginal osteophytes"	Small bony spurs at the edges of the joint. Very common from middle age.	Nothing on its own. Spurs do not cause the pain; the joint process does.
"Joint space narrowing, medial compartment"	The inner side of the knee has less cartilage space than normal. The most common pattern in Indian knees.	Strengthening, weight work, and a check on whether you are becoming bow-legged.
"Subchondral sclerosis"	The bone under the cartilage has thickened in response to load.	Confirms osteoarthritis. Changes nothing about the first-line plan.
"Subchondral cysts"	Small fluid-filled spaces in the bone beneath the cartilage.	A marker of more advanced change. Still treated by symptoms.
"Varus deformity"	The knee is drifting bow-legged, loading the inner side more.	Worth a proper conversation about bracing, insoles and the timeline for surgery.
"Joint effusion"	Fluid in the joint, i.e. it is swollen.	This is a flare. Chapter 19 has the 48-hour plan.
"Baker's cyst"	A fluid-filled sac behind the knee, almost always a symptom of fluid in the joint itself.	Treat the knee. Do not let anyone operate on the cyst.
"Chondrocalcinosis"	Calcium deposits in the cartilage, suggesting pseudogout.	Worth mentioning to your doctor, especially with sudden hot swollen attacks.
"Degenerative changes"	The radiologist's shorthand for osteoarthritis of some grade.	Read the grade. Then read your own symptoms. Chapter 6 decides what happens next.

Two practical habits. First, always get the images, not just the report: films or a disc, plus the report, and keep them. Indian patients routinely change cities, hospitals and doctors, and a repeat X-ray in six months is easy to avoid if the original is in your folder. Second, ask the orthopaedic surgeon to show you the film on the screen and point at what they are describing. A surgeon who will not do that is telling you something.

When the report and the pain disagree

They disagree often, and the pain usually wins. A report is a description of structure at one moment. Your pain is a description of function over weeks. If you have grade 3 changes and can walk two kilometres, you are not a surgical candidate on the basis of the film. If you have grade 2 changes and cannot get off a low chair, you need treatment now regardless of the grade.

This is also why a second opinion is not an insult to your doctor. It is a second reading of a decision with a ten-to-twenty-year consequence. Chapter 15 gives you the specific questions to ask, and the four situations in which a second opinion is close to mandatory.

6. The evidence ladder: what works, what is optional, what to skip

Twenty-eight things Indians buy for knee pain, sorted by what the guidelines and trials actually say. Green earns its place. Amber is a judgement call. Red is money you will not get back.

This table is built from four guideline documents that agree with each other more than they disagree: the ICMR Standard Treatment Workflow for knee osteoarthritis (July 2024), NICE guideline NG226 (October 2022), the OARSI non-surgical management guidelines (2019) and the American College of Rheumatology's 2019 osteoarthritis guideline. Where they conflict, the conflict is noted.

Treatment	Verdict	What the evidence says
Education and self-management	Green	A core treatment in OARSI and NICE. Knowing what the condition is, what a flare means, and what to do reduces pain and visits.
Structured strengthening exercise	Green	A core treatment in every guideline, and named first in the ICMR workflow: quadriceps strengthening and hamstring stretching.
Aerobic exercise (walking, cycling)	Green	Recommended alongside strengthening by NICE. Improves function and reduces pain.
Weight loss, 5-10% of body weight	Green	NICE: any loss helps; 10% is likely better than 5%. Each kilogram removed takes roughly four off each step.
Topical NSAID gel	Green	Strongly recommended for knee OA by OARSI (Level 1A) and offered by NICE. Good pain relief with far less stomach and kidney exposure than tablets.
Walking stick, used correctly	Green	Cheap, effective, under-prescribed. Held in the hand opposite the painful knee.
Heat and cold	Green	Not disease-modifying, but useful for comfort and for getting through a flare. Cold for a hot swollen knee, heat for stiffness.
Supervised physiotherapy	Green	NICE suggests considering supervised sessions. In India this is often the difference between a programme that happens and one that does not.
Tai chi and gentle yoga	Green (moderate)	A 2024 meta-analysis of 8 trials and 756 patients found yoga reduced pain (effect size -0.92), stiffness (-0.51) and improved function (-0.53). Avoid deep kneeling postures.
Aquatic exercise	Amber	OARSI rates it 1B/2 depending on comorbidity. Excellent if you are heavy or have other joint pain. Needs a pool, which is the practical barrier in most Indian cities.
Paracetamol	Amber	OARSI conditionally recommends against it for knee OA; NICE does not offer it routinely. It is safe and cheap, and it is not a strong painkiller for this condition.
Oral NSAIDs, short course	Amber	They work. They also carry stomach, kidney, blood pressure and cardiovascular risk. Lowest dose, shortest time, never as a permanent habit.
Intra-articular steroid injection	Amber	Good short-term relief, especially during a flare with fluid. NICE puts the relief window at 2 to 10 weeks. Not a repair. Three to four in the same knee per year is the accepted ceiling.
Hyaluronic acid injection	Amber	The guidelines disagree: OARSI rates it a conditional option (1B/2) and EULAR calls it probably effective with a small effect size, while NICE says do not offer it at all. Expensive for a modest, temporary benefit.
Duloxetine	Amber	Recommended by some guidelines for chronic musculoskeletal pain. A prescription decision, not a self-help one.
Knee brace	Amber	NICE says do not routinely offer braces, but allows them for instability or abnormal loading when exercise alone is not enough. Not a substitute for strength.

Treatment	Verdict	What the evidence says
Insoles and footwear change	Amber	Cushioned shoes help symptoms. NICE does not routinely offer insoles; custom ones are rarely worth the money for osteoarthritis alone.
Acupuncture and dry needling	Red	NICE says do not offer acupuncture or dry needling to manage osteoarthritis. The evidence of benefit is moderate to very low quality.
Electrotherapy: TENS, ultrasound, laser	Red	NICE says do not offer any of these for osteoarthritis because the evidence of benefit is insufficient. Indian clinics still sell them by the session.
GLP-1 injections (semaglutide)	Amber	Real effect in obesity with knee OA, shown in a 2024 trial. A prescription medicine with a specific population, not a knee drug.
Approved MSC therapy (Ciplostem)	Amber	DCGI-approved, single-dose, for grade 2 and 3. Narrow indication, real cost, ask hard questions. Chapter 12.
Glucosamine and chondroitin	Red	The American College of Rheumatology strongly recommends against glucosamine, alone or with chondroitin. A 2018 analysis of 29 studies found inconsistent results.
Collagen supplements	Red	Small, short trials with mixed results. No major guideline recommends collagen for knee pain.
Turmeric and curcumin capsules	Red	Small trials, frequently industry-funded, no guideline recommendation. Turmeric in food is fine; capsules are a cost, not a treatment.
Oral and transdermal opioids, including tramadol	Red	OARSI strongly recommends against them (Level 5). Falls, dependence, seizures, interactions.
PRP injections	Red	No supportive guideline recommendation, and under active regulatory scrutiny in India. Chapter 12 has the details.
Arthroscopic lavage or debridement for OA	Red	Does not help established osteoarthritis. Major guideline bodies advise against routine use for this indication.
"Cartilage regrowth" clinics and unapproved cell injections	Red	India's regulatory framework for unapproved cell therapies is widely described in the legal literature as ambiguous and poorly enforced. Chapter 12.

How to use this table in a consultation

Do not argue with your doctor using a table from a book. Ask instead: "Of the things that work, which two should I start this month?" A good clinician will pick strengthening and weight, and add a topical gel. If the answer is an injection and a supplement, ask what the plan is for the underlying problem.

7. The eight-week strength programme that does the actual work

The ICMR's conservative treatment for knee osteoarthritis is one sentence long: quadriceps strengthening and hamstring stretching, lifestyle modification, weight loss. This chapter is that sentence, made doable in an Indian living room with a chair, a wall and a towel.

Strength is not a wellness idea here. It is the mechanism by which the joint tolerates load. The quadriceps absorbs force before it reaches the knee; the hamstrings and glutes control how the knee tracks; the calf and hip muscles decide how you walk. Weakness in any of these turns an ordinary walk into a series of small injuries.

The rules, before the exercises

- **Two to three sessions a week, not seven.** Muscle adapts during recovery. Daily hard sessions in a painful knee produce a flare, not strength.
- **Pain up to 3 out of 10 during exercise is acceptable.** Sharp pain, pain that makes you limp, or pain that is still worse 24 hours later is not.
- **Settle within a day.** The test is the next morning. If the knee is more swollen or more painful the day after, you did too much. Reduce by a third and rebuild.
- **Progress by two reps first, then by resistance.** Not by adding a new exercise.
- **Never work through a hot, swollen knee.** During a flare, do only the isometric quad sets and ankle pumps until the swelling settles.
- **Breathe.** Holding your breath raises blood pressure and makes the exercise harder than it needs to be.

The twelve exercises

Exercise	How to do it	Dose	What it is for
1. Quad set	Sit with the leg straight. Press the back of the knee down into the bed, tighten the thigh, hold 5 seconds.	10 reps, twice daily	Wakes up the muscle without loading the joint. Safe even on a flare day.
2. Straight leg raise	Lie down, one knee bent. Tighten the straight thigh and lift the leg 20 cm, hold 3 seconds.	10 reps, each leg	Quadriceps without knee movement.
3. Seated knee extension	Sit on a chair. Straighten one knee fully, hold 5 seconds, lower slowly.	12 reps, each leg	The inner quadriceps, which wastes fastest.
4. Sit-to-stand	From a chair with arms, stand up without using your hands, sit slowly.	8-12 reps	The single most useful daily-life strength test and exercise.
5. Wall sit	Back against a wall, slide down until knees are at about 60 degrees. Never below 90.	3 holds of 20-30 seconds	Builds endurance in the quadriceps with low joint stress.
6. Step-up	A 10 cm step, then 15 cm. Step up leading with the affected leg, step down leading with the other.	10 reps, each leg	Trains the pattern of stairs, which is where most pain appears.
7. Heel slide	Lie down, slide the heel towards the buttock as far as comfortable, hold 5 seconds.	10 reps	Restores bending, which stiffness steals.
8. Hamstring curl	Stand holding a chair. Bend the knee to bring the heel towards the buttock, slowly.	12 reps, each leg	The ICMR's hamstring work. Protects the knee from behind.
9. Calf raise	Stand, rise onto the toes, hold 2 seconds, lower slowly.	15 reps	Ankle and calf power for walking and balance.

Exercise	How to do it	Dose	What it is for
10. Glute bridge	Lie on your back, knees bent, lift the hips, squeeze the buttocks, hold 3 seconds.	12 reps	Hip strength reduces load transmitted to the knee.
11. Side-lying leg lift	Lie on your side, lift the top leg 30 cm, keep the toes forward.	12 reps, each side	Gluteus medius. Weakness here makes the knee drift inward.
12. Single-leg balance	Stand on one leg near a wall, 20-30 seconds. Progress by closing your eyes only if safe.	3 rounds each leg	Balance and confidence, which prevent falls.

How the eight weeks progress

Weeks	What you do	What should change
1-2	Quad set, straight leg raise, seated extension, heel slide, calf raise. Two sessions a week, plus a short daily walk.	Confidence. Pain may briefly increase; it should settle within 24 hours.
3-4	Add sit-to-stand and hamstring curl. Start wall sits at 30 degrees. Three sessions a week.	Stairs become easier. Standing up from a low chair is less of an event.
5-6	Add step-ups, glute bridge and side-lying lift. Wall sits to 60 degrees. Resistance band on the seated extension.	Walking distance increases. You stop planning around the knee.
7-8	Add single-leg balance. Increase step height to 15 cm. Two heavier sets instead of three light ones.	Measurable change: more sit-to-stands in 30 seconds, longer single-leg stand, fewer pain days.

Equipment, in total: a chair with arms, a wall, a rolled towel, a 1-litre water bottle as a weight, and optionally a resistance band for Rs 200-400. The step can be the bottom stair, a thick book, or a stable plastic stool. Nothing in this programme requires a gym, and none of it requires you to leave the house.

What to stop doing

- Deep squats and full lunges to the floor. They load the patellofemoral joint at the angle where it has least contact area.
- Full lotus or long virasana holds. If you cannot get down and up without using your hands, do not get down.
- Stair climbing as exercise. Stairs are a functional necessity; they are a poor training choice for a painful knee.
- Running on concrete as your first activity. Build strength first, then distance, then speed.
- "No pain, no gain" in a joint with inflammation. Pain during exercise is information, not a badge.

> **The three signs it is working**
> You can stand up from the floor without using your hands. You take the stairs without thinking about which leg to lead with. Your pain-free walking distance has doubled. None of these shows on an X-ray, and all three are worth more than a grade.

8. Walking, cycling, swimming and yoga: how much, how hard

Strength is the foundation. Movement is the medicine that keeps the whole system working. The dose matters more than the activity you choose.

Activity	Starting dose	How to progress	Watch out for
Walking	10 minutes twice a day on flat ground.	Add 5 minutes a week to one session until you reach 30 minutes most days. Pace: you can hold a conversation.	Pain that builds over the walk and is still there next morning. Concrete pavements, worn slippers, long downhill stretches.
Stationary cycling	10 minutes, low resistance, seat high enough that the knee never fully straightens at the bottom.	Add 5 minutes a week, then resistance. Two minutes easy, one minute harder.	Seat too low, which forces deep knee bend under load. Sweating in a hot room without water.
Swimming and pool work	20 minutes, walking in chest-deep water if you cannot swim.	Add flutter kick with a board, then full laps.	Breaststroke kick, which twists the knee. Getting in and out of the pool safely.
Yoga	Two sessions a week, 20-30 minutes, avoiding deep kneeling.	Add balance postures and standing strength postures.	Virasana, padmasana, malasana and long holds in deep flexion. A teacher who pushes you into them.
Gym	Leg press with a limited range (never below 90 degrees of knee bend), light resistance.	Add 5% resistance weekly. Add hip abduction and calf work.	Deep squats under load, leg extension machines at heavy weight, treadmill incline walking as a first activity.
Tai chi	10-minute form, twice weekly.	Full 20-minute form.	Very low stances in the first weeks.

The 10% rule keeps you out of trouble: never increase weekly volume by more than about a tenth. A knee that tolerated 60 minutes last week will usually tolerate 66 this week. It may not tolerate 120, which is what people do when they feel better and decide to catch up on three months of lost walking in a weekend.

Yoga deserves its own paragraph

Yoga is the most popular exercise prescription in India and, unusually, it has decent evidence behind it for this condition. A 2024 systematic review and meta-analysis in PLOS ONE pooled eight randomised trials covering 756 patients and found statistically significant improvements in pain (standardised mean difference -0.92), stiffness (-0.51) and physical function (-0.53). It did not find significant improvement in quality of life. That is a real effect on the things that hurt, and no effect on the things that are measured with questionnaires.

The practical problem is that the postures most associated with Indian practice are the ones that hurt this joint. Sitting in vajrasana for twenty minutes is a sustained deep knee bend. Padmasana loads the inner compartment and the menisci. A good teacher will give you a modified sequence: standing postures, supported balance, gentle twists, no prolonged kneeling. If your teacher's response to knee pain is "do more, your flexibility is poor", find another teacher.

Choosing a physiotherapist in India

A physiotherapist who gives you a printed programme, teaches each exercise, checks your form once a week and discharges you with a home plan is worth ten sessions of passive treatment. One who puts you on a heat pack, does ultrasound for fifteen minutes and hands you a bill is not. Ask at the first visit: what is my diagnosis, what is the programme, how many sessions, and what will I be able to do at the end that I cannot do now? The answers tell you everything.

9. Weight: the arithmetic of a lighter knee

Losing weight is the only intervention in this book that changes the load on the joint, the inflammation in the body and the risk of diabetes at the same time. It is also the one most reliably sabotaged by vague advice.

The best-known number in knee osteoarthritis research comes from a Wake Forest study published in 2005. For every pound of body weight lost, the compressive load on the knee fell by about four pounds per step. The researchers calculated that a one-pound loss meant more than 4,800 pounds less load per mile walked, and that ten pounds lost meant each knee carried 48,000 pounds less compressive load for every mile. In metric terms, each kilogram you lose takes roughly four kilograms off each step you take.

That is why the guidelines are so direct. NICE tells clinicians to explain that any amount of weight loss is likely to be beneficial, and that losing 10% of body weight is likely to be better than 5%. A 2013 randomised trial, known as IDEA, found that diet plus exercise produced greater improvement in pain and function than either alone, along with a larger fall in inflammatory markers. Neither diet nor exercise alone matched the combination.

Use the Indian cut-offs, not the Western ones

The ICMR-National Institute of Nutrition's Dietary Guidelines for Indians, revised in 2024, use Asian BMI cut-offs because Indians carry metabolic risk at lower body weights. If you have been told your BMI is "normal" by an international calculator, check it against this table.

Measure	Indian cut-off	Why it matters for your knee
BMI under 18.5	Underweight	Muscle loss matters here. Strength work and protein, not calorie cutting.
BMI 18.5 to 22.9	Normal	Protect it. This is the range where the knee load is lowest.
BMI 23 to 24.9	Overweight	The range where Indian guidelines say risk begins. A 5% loss is a realistic target.
BMI 25 and above	Obese	A 10% loss is the target that NICE suggests is worth aiming for.
Waist 90 cm or more (men)	Abdominal obesity	Visceral fat drives inflammation. Measure at the navel, standing, after breathing out.
Waist 80 cm or more (women)	Abdominal obesity	Same measure, same meaning.

For context: NFHS-5 data summarised in the same ICMR-NIN guidelines show overweight or obesity in 22.9% of men and 24.0% of women aged 18 to 69, with abdominal obesity far higher at 47.7% of men and 56.7% of women on NNMB data. Thin arms and a large waist is the commonest Indian body shape, and the waist is what matters here.

Protein, without the powder

The ICMR-NIN 2024 guidelines set the protein requirement for an average adult at 0.83 g per kilogram of body weight per day, and note that a cereal-based Indian diet needs about 1 g per kilogram to compensate for lower digestibility. For a 70 kg adult that is roughly 58 to 70 grams a day. The same guidelines explicitly advise obtaining protein from a combination of foods and avoiding protein supplements to build muscle.

Practically, that means dal at two meals, a bowl of curd, paneer or eggs, and a portion of chana, rajma or soya at least a few days a week. It does not mean a Rs 3,000 tub of whey. Protein matters during weight loss because it protects the muscle you are trying to build in Chapter 7, and losing muscle while losing fat makes a knee worse, not better.

A 1,600-calorie Indian day that a family can actually cook

Calorie counts are approximations. What matters is the structure: protein at every meal, vegetables filling half the plate, one grain portion at each meal rather than two, and no liquid sugar.

Meal	What it looks like	Why
Breakfast	Two medium rotis (60 g atta) with a bowl of moong dal or two eggs, plus a cup of curd or a glass of milk.	Protein early stops the mid-morning biscuit.
Mid-morning	A fruit and a handful of roasted chana. Tea without sugar, or with half a spoon.	Fibre and protein, no packet.
Lunch	One cup of rice or two rotis, one bowl of dal or rajma, a large bowl of sabzi, salad, curd.	Half the plate is vegetables. Grain is one portion, not the meal.
Evening	Sprouts or a small bowl of peanut chikki instead of namkeen. Water before tea.	The 5 pm slot is where most Indian diets add 400 empty calories.
Dinner	Two rotis or a small bowl of rice, paneer, fish, chicken or soya, two vegetables, no second helping.	Earlier dinner, lighter grain, protein kept.
What to cut	Sugar in tea, fried snacks, sweetened drinks, restaurant food more than once a week, alcohol.	These are the calories that come with no protein and no fibre.

Expected rate: 0.5 to 1 kg a week at the start, slowing to 0.25 to 0.5 kg. A 5% loss in six months is a good result and a meaningful one for the knee; 10% over a year is better. Anyone with diabetes, kidney disease or on multiple medicines should do this with a doctor or dietitian rather than from a book, and anyone who has had an eating disorder should not use a calorie table at all.

The number that will keep you going

Lose 5 kilograms and each step you take carries about 20 kilograms less. You take roughly 5,000 to 7,000 steps on an ordinary day. That is over 100,000 kilograms of load removed from one knee, every single day, without an operation, an injection or a supplement.

10. Your Indian home is a gym with bad ergonomics

You do not need a new house. You need six changes, most of them under Rs 2,000, that remove the deepest and most repeated knee bends of your day.

The Indian household loads the knee in ways a Western guideline never considers. Floor seating for meals, an Indian-style toilet, washing clothes by hand, sweeping and mopping, cooking at a low platform, praying while kneeling, and stairs in buildings without lifts. The Tiruvallur study found Indian toilet use associated with more than twelve times the odds of knee osteoarthritis, the second-largest association it measured after menopause. Whatever the causal debate, the ergonomics are obvious: it is the deepest squat most people perform, several times a day, often with nothing to hold.

Situation	The problem	The swap that works
Indian-style toilet	A full squat, repeated daily, with nothing to hold and no way to control the descent.	A commode stool or raised seat frame over the pan (Rs 800-2,000), plus a wall-mounted grab bar. If you cannot install one, a plastic stool to sit on with the pan used as a floor drain. This is the highest-value purchase in this book.
Floor seating at meals	Cross-legged sitting loads the inner compartment and is hard to get out of.	A low chair or a folding stool (Rs 500-1,200). Keep one folded near the dining area. Use it on flare days without apology.
Getting up from the floor	The most dangerous movement for a painful knee, and a common cause of falls.	Roll onto your side, push up with the arms onto hands and knees, then bring one foot forward and use a chair. Never force a straight-up rise.
Stairs at home	Every trip loads the patellofemoral joint, especially going down.	Up with the good leg first, down with the bad leg first. Hold the rail. Combine trips. If there is a bedroom downstairs, use it during a flare.
Mopping and sweeping	Sustained squatting and twisting under load.	A long-handled mop, mopping in standing position, and a stool for any floor work. Delegate during flares.
Cooking at a low platform	Long periods in a squat or on a low stool.	A height-adjustable stool, or a small chair. Batch cooking on good days.
Prayer and namaz	Kneeling and deep flexion, sometimes for long periods.	A chair is acceptable in most traditions for those who cannot kneel. Ask your religious authority; almost all permit it for the ill.
Two-wheeler and car	Mounting and dismounting in flexion, plus long static sitting.	Enter the car seat-first, then swing both legs in. On a two-wheeler, mount from the left with the stand down. Break long drives every 90 minutes.
Metro and bus	Standing in a moving vehicle, stairs at stations.	Use the lift where available, board at a door near a seat, and carry a folded stick if balance is uncertain.
Footwear	Flat rubber chappals and worn soles transmit impact; high heels shift load to the kneecap.	Cushioned walking shoes with a firm heel counter for anything over ten minutes. Replace when the sole is worn at the outer edge.

Braces, sticks and other hardware

A walking stick held in the hand opposite the painful knee reduces load on that knee and costs a few hundred rupees. It is the most under-used device in Indian orthopaedics, largely because of how it looks. A single upright stick with a rubber tip, set at the height of your wrist crease when standing, is all you need.

Braces are more complicated. A simple sleeve brace provides warmth and confidence and little else. An unloader brace can shift load away from one compartment, which helps some people with single-compartment disease, but it is bulky and expensive and should be fitted by someone who knows what they are doing. A hinged brace is for instability, not for pain. None of them replaces strength, and none of them should be worn all day for months.

> **The Rs 3,000 knee kit**

A commode stool or raised seat (Rs 800-2,000), a grab bar (Rs 400-800), a resistance band (Rs 200-400), a walking stick (Rs 300-600), and a folding stool for floor seating (Rs 500-1,200). That is the entire equipment budget for the non-surgical treatment of knee osteoarthritis, and it will do more than a year of glucosamine.

11. Painkillers without wrecking your kidneys

The order of operations is settled: gel first, tablets second, short courses only, and opioids almost never. Most Indian knee patients are doing the reverse.

OARSI strongly recommends topical NSAIDs for knee osteoarthritis (Level 1A). NICE tells clinicians to offer a topical NSAID to people with knee osteoarthritis. Both are telling you the same thing: the drug that reaches the knee through the skin gives useful relief with a small fraction of the systemic exposure that a tablet delivers. If you have not tried a gel properly, for two weeks, four times a day, you have skipped the best-value treatment in the pharmacy.

The gel, done properly

- Two to four grams, which is roughly a two-to-four centimetre strip, four times a day, rubbed into the painful part of the knee. Not once a day when it hurts.
- Wash your hands afterwards. Do not apply it and then put a hot water bag or heating pad on the knee.
- Do not use a gel and a tablet of the same drug class on the same day without asking. You are doubling the dose, not adding a treatment.
- Give it two weeks of consistent use before deciding it does not work.
- Avoid on broken skin, and check with your doctor if you have kidney disease, heart failure or are on blood thinners.

Tablets: what each one actually does

Medicine	Real role	The catch
Paracetamol	Cheap, safe at label doses, weak for osteoarthritis pain. OARSI conditionally recommends against it for knee OA.	Liver risk in overdose. It is in dozens of combination cold and pain products; never take two products containing it.
Diclofenac, aceclofenac, ibuprofen, naproxen	Genuine pain relief for a flare, at the lowest dose for the shortest time.	Stomach ulcers and bleeding, raised blood pressure, kidney injury, cardiovascular risk. Take with food. Not for long-term daily use without review.
Etoricoxib and other COX-2 inhibitors	Similar relief with less stomach risk, useful if you have had an ulcer.	Cardiovascular risk. OARSI advises against any oral NSAID in people with cardiovascular disease or frailty.
A proton pump inhibitor (omeprazole, pantoprazole)	Protects the stomach if you must take an NSAID and are at risk.	Not a licence to take NSAIDs indefinitely. OARSI pairs NSAID plus PPI at Level 2 for people with gastrointestinal risk.
Tramadol	Sometimes used when nothing else is possible.	OARSI strongly recommends against opioids for knee OA. Seizures, falls, drowsiness, interactions with antidepressants. Not a "safe weak" option.
Muscle relaxants and combination pain products	Widely sold in India for knee pain.	Sedation, no evidence for osteoarthritis, and often contain paracetamol you are already taking.

Who should not take an oral NSAID without a conversation

- Anyone with reduced kidney function, heart failure, or uncontrolled blood pressure.
- Anyone with a previous stomach or duodenal ulcer, or a history of gastrointestinal bleeding.
- Anyone on blood thinners, aspirin for the heart, or clopidogrel.
- Anyone on an ACE inhibitor or angiotensin receptor blocker plus a diuretic, a combination that with an NSAID is a well-known cause of acute kidney injury.

- Anyone in the third trimester of pregnancy.
- Anyone over 75, who should be on the lowest dose for the shortest possible time, with stomach protection considered.

None of this is a reason to suffer. It is a reason to use the gel, use the tablets for three to five days at a time, and put the effort into the things that change the trajectory: strength, weight and load management. A knee that is well supported by muscle needs fewer painkillers, which is the outcome you actually want.

> **The medicine cabinet, sorted**
> Keep: a topical NSAID gel, paracetamol for occasional use, and a cold pack. Ask before adding: any oral NSAID, any combination product, any supplement. Discard: expired tablets, old antibiotics, and anything a chemist recommended without a diagnosis.

12. Injections: what a needle can and cannot do

An injection buys time. It does not rebuild cartilage, and in India it is also the most aggressively marketed product in the entire knee market.

There are four things that get injected into knees in India: corticosteroid, hyaluronic acid, platelet-rich plasma, and cells. They are not interchangeable, and only one of them has a price cap, a clear indication and a guideline behind it.

Corticosteroid: the honest version

A steroid injection reduces inflammation in the joint lining. It works fastest and best when there is a flare with fluid on the knee; the European alliance of rheumatology associations (EULAR) recommends it for an acute exacerbation of knee pain, particularly when accompanied by an effusion. Relief typically peaks in the first couple of weeks and fades over weeks to a few months. It is not a repair, and repeated injections do not accumulate benefit.

- **How many is too many?** EULAR notes that avoiding more than three to four steroid injections in the same joint per year is a generally accepted rule, while being explicit that this rule is based on no research evidence. The American College of Rheumatology advises no more often than every three months.
- **Infection risk:** septic arthritis after an intra-articular injection occurs in roughly one in 14,000 to one in 77,000 procedures. Rare, serious, and the reason technique and a clean setting matter.
- **If you have diabetes:** blood sugar can rise for two to three days afterwards. Plan for it, test more often, and tell the doctor who manages your diabetes.
- **Skin changes:** thinning and lightening of the skin at the injection site are more visible on darker skin. Ask about it before, not after.
- **Cost in India:** typically a few thousand rupees including the doctor's fee, which is why it feels like good value for a wedding or a long trip. It is poor value as a monthly habit.

Hyaluronic acid: small effect, real money

Hyaluronic acid is injected to supplement the joint's natural lubricant. OARSI rates it a conditional option (Level 1B or 2 depending on other conditions). EULAR's summary is fair: probably effective in knee osteoarthritis, but the size of the effect is relatively small, the patients who benefit most are not well defined, and the economics are not established. Courses are usually three to five injections. At Indian private prices that is a meaningful sum for a modest, temporary benefit, and it is not covered by most insurance.

PRP: the market is ahead of the evidence

Platelet-rich plasma is made from your own blood, spun down and injected. It is marketed heavily in Indian cities as cartilage regeneration. In March 2025 the Telangana Medical Council began reviewing allegations about PRP therapy marketing after complaints from 75 orthopaedic surgeons, with reported charges of Rs 5,000 to Rs 10,000 per injection and packages running into lakhs. The Telangana Orthopaedic Surgeons Association publicly cautioned patients against unproven claims.

No major guideline recommends PRP for knee osteoarthritis. The trials are small, heterogeneous and inconsistent. The claim that it regrows cartilage is not supported. If a clinic is selling you a Rs 1 lakh course of four injections on the promise of avoiding surgery, ask for the published trial that supports that promise, in that population, with that outcome.

Stem cells: one approved product, and a lot of everything else

This is the part of the market where the difference between a legitimate therapy and a scam is largest. In December 2025 Cipla launched Ciplostem, an allogeneic mesenchymal stromal cell therapy approved by the Drug Controller General of India, developed by Stempeutics. It is a single intra-articular dose of 25 million bone-marrow-derived mesenchymal stem cells given with hyaluronic acid, indicated for grade 2 and grade 3 knee osteoarthritis. The company states that it reduces pain and preserves function for up to two years after administration.

That is a real, regulated product with a phase 3 programme behind it, and it is not the same thing as the injection being sold in a small clinic from a centrifuge in the back room. A 2025 analysis in the Journal of Law and the Biosciences describes India's regulatory framework for cell therapies as ambiguous, poorly implemented and associated with a rise in unproven stem cell offerings. Both things are true at once.

If you are considering cell therapy, five questions decide it:

1. Is this the DCGI-approved product, with a brand name, a batch number and a package insert?
2. Am I grade 2 or 3 on a standing X-ray? It is not indicated outside that group.
3. What is the total cost, in writing, including follow-up?
4. What does the published evidence say about how long the effect lasts, in patients like me?
5. Who is injecting it, and what is their complication rate?

> **The five-question injection test**
> What is in the syringe? What is the evidence it works for my grade? How long does the effect last? What are the risks in my body, with my other conditions? What is the plan if it does not work? Any clinic that cannot answer all five in plain language is selling a procedure, not a treatment.

13. GLP-1 injections and your knee: a real effect with real caveats

In 2024 a randomised trial showed that semaglutide reduced knee osteoarthritis pain substantially in people with obesity. It is the most important piece of knee research in a decade, and it is being oversold to people it was not tested on.

The trial, published in the New England Journal of Medicine in October 2024 and known as STEP 9, enrolled 407 adults with a BMI of 30 or more and moderate knee osteoarthritis with at least moderate pain, across 61 sites in 11 countries. Participants received weekly semaglutide 2.4 mg or placebo for 68 weeks, and both groups got counselling on physical activity and a reduced-calorie diet.

Outcome at 68 weeks	Semaglutide	Placebo	What it means
Change in body weight	-13.7%	-3.2%	Large, real weight loss. The drug did what it was designed to do.
Change in WOMAC pain score (0-100, higher is worse)	-41.7 points	-27.5 points	Both groups improved a lot. The drug added about 14 points on top of diet and activity advice.
SF-36 physical function	+12.0 points	+6.5 points	Better function, not just less pain.
Stopped treatment due to side effects	6.7%	3.0%	Mostly gastrointestinal. This is a drug with real side effects.

Three caveats that the marketing leaves out

1. **The trial population was not the typical Indian knee patient.** Mean BMI was 40.3 and 81.6% of participants were women. Most Indian patients with knee osteoarthritis sit between BMI 25 and 32. The drug has not been tested at that weight for this outcome, and the effect there is likely to be smaller.
2. **Placebo plus counselling did very well.** The placebo group lost 3.2% of body weight and improved 27.5 points on pain. Diet advice and an activity programme are not a control condition in real life; they are a treatment, and a free one.
3. **It is a treatment for obesity and diabetes, not a knee drug.** It is prescription-only, needs monitoring, and stopping it usually means regaining the weight, which means the knee load comes back.

If you have obesity and knee osteoarthritis, this is a legitimate conversation to have with an endocrinologist or physician, especially if you have diabetes or prediabetes as well. Several brands of semaglutide are now sold in India and prices vary widely by brand and dose, so ask your doctor and chemist for current pricing rather than trusting any figure in print, including this one. What you should not do is buy it from an online seller without a prescription, or expect it to replace the exercise programme.

14. The supplement habit, audited

The average Indian knee patient spends more each month on supplements than on physiotherapy. Almost none of it has guideline support. Here is the audit, with what to do with the money instead.

Product	The claim	What the evidence says	Verdict
Glucosamine, with or without chondroitin	"Rebuilds cartilage."	The American College of Rheumatology strongly recommends against glucosamine, alone or combined. A 2018 analysis of 29 studies and 6,120 participants found inconsistent results across trials.	Skip. This is the largest single waste in the category.
Collagen peptides	"Feeds your cartilage."	Small, short trials with mixed results. No major guideline recommends it for knee osteoarthritis.	Skip unless you have money to spare and accept the uncertainty.
Turmeric or curcumin capsules	"Natural anti-inflammatory."	Small trials, often industry-funded, with inconsistent results. No guideline recommendation.	Skip the capsules. Cook with turmeric; it costs nothing and tastes better.
Omega-3 fish oil	"Reduces joint inflammation."	Trials in knee osteoarthritis have not shown meaningful pain benefit. Reasonable for other reasons, not for this one.	Not a knee treatment.
Vitamin D	"Strengthens the joint."	Correcting a deficiency is worthwhile for bone and muscle. Trials of vitamin D supplementation for knee osteoarthritis pain and cartilage loss have been negative.	Test first. Treat a deficiency. Do not expect knee pain relief.
Calcium	"For the bones."	Relevant to bone density and fracture risk, not to cartilage or knee pain.	Only if your doctor advises it.
Ayurvedic churna and proprietary tablets	"Removes stiffness at the root."	Little high-quality trial evidence for osteoarthritis outcomes. Quality control varies between manufacturers.	If you want to try it, buy from a licensed pharmacy with a batch number, and tell your doctor, because some preparations interact with other medicines.
Protein powder	"Builds muscle to support the knee."	The ICMR-NIN 2024 guidelines explicitly advise obtaining protein from foods and avoiding protein supplements to build muscle.	Food first. Dal, curd, eggs, paneer, soya.

A household spending Rs 2,500 a month on glucosamine and collagen spends Rs 30,000 a year. That is six to eight supervised physiotherapy sessions, a good pair of walking shoes, a commode stool, a grab bar, a resistance band, a year of gym membership, and still enough left over for a decent pair of insoles. The supplements do not have a mechanism that survives contact with a trial. The other things do.

One exception worth making

If a blood test shows you are deficient in vitamin D or B12, treat it. That is not a supplement habit, it is correcting a documented deficiency, and it affects muscle strength, balance and falls. The rule is simple: test, then treat. Not treat, then hope.

15. Is surgery actually needed?

The ICMR document that Indian orthopaedic surgeons work from carries a line across the top: keep a high threshold for invasive procedures. It then lists exactly when a knee replacement is justified, and the list is narrower than most patients expect.

According to the ICMR Standard Treatment Workflow for osteoarthritis of the knee (July 2024), patients with Kellgren-Lawrence grade 4 disease may require surgery after appropriate conservative treatment has failed, when they have:

- Severe knee pain or stiffness that limits activities of daily living, including walking, climbing stairs, and getting in and out of chairs.
- An inability to walk short distances because of significant knee pain, requiring a cane or walker.
- A regular analgesic requirement over a long period.
- Moderate or severe knee pain at rest, either during the day or at night.
- Severe deformity.

It also notes that a high tibial osteotomy may be considered in younger patients with significant varus deformity, which is a real alternative that many Indian patients are never told about. And it lists contraindications to knee replacement: recent or current knee infection, a remote source of ongoing infection, discontinuity or severe dysfunction of the extensor mechanism, recurvatum deformity from neuromuscular weakness, and a painless, well-functioning knee fusion.

"Failed conservative treatment" has a definition, and most people never meet it

The phrase is doing enormous work in that guideline, and in practice it is often used to mean "the patient came back". A fair trial of conservative treatment looks like this: at least twelve weeks of a structured strengthening programme, done at least twice a week; a genuine attempt at weight loss if you are overweight; a properly used topical NSAID; and pain control that has been optimised rather than abandoned. If you have not had that, you have not failed conservative treatment. You have not had it.

Eight questions for the surgeon

Question	What a good answer sounds like
Which compartment is affected, and what grade am I?	A specific answer with the film in front of you, pointing at the joint line.
What will I be able to do after surgery that I cannot do now?	Concrete: walk to the market, climb stairs, sleep without waking. Not "better quality of life".
What happens if I wait six months?	An honest answer about whether the delay carries risk. Usually it does not, and that is useful to hear.
How many of these do you do a year, and what is your complication rate?	A number. Reluctance to answer is itself an answer.
Which implant will you use, and is it under the NPPA price cap?	A brand, a model, and an invoice line. See Chapter 16.
What is the all-in cost, including physiotherapy and one year of follow-up?	A written estimate, not a per-day room rate.
What is my plan if the pain is not better at six months?	A follow-up plan that exists before the operation.
Who manages my rehabilitation, and when does it start?	A named physiotherapist, starting on day one.

Partial or total, and what "robotic" means

A unicompartmental (partial) replacement replaces only the worn side of the knee. It works well in a narrow group: disease confined to one compartment, a good range of movement, a functioning anterior cruciate ligament, and a patient who is not significantly obese. It recovers faster and feels more like a normal knee, but it has a higher revision rate and is offered far less often in India than in the West.

Robotic and computer-assisted surgery improve the accuracy of alignment and reduce the number of badly aligned knees. That is well established. Whether it produces better function at ten years is not, and the premium of one to two lakh rupees is real. If a hospital is selling robotics as the reason your knee will last longer, ask for the study that shows it in patients like you.

What does not work for osteoarthritis is arthroscopy: washing out or shaving the joint. Major guideline bodies advise against routine arthroscopic lavage or debridement for established knee osteoarthritis, because it does not improve pain or function beyond placebo. If you are offered a "keyhole cleaning" for arthritis, ask what it is meant to achieve that exercise has not.

Nobody should be refused on the basis of age or BMI alone

NICE is unusually blunt about this. It says clinicians should not exclude people with osteoarthritis from referral for joint replacement because of age, sex or gender, smoking, comorbidities, or overweight and obesity measured by BMI, because the risks of the operation vary with these factors rather than making the operation impossible. It also says the decision to refer should be made on clinical assessment, not on a numerical score of disease severity.

This matters in India, where a patient at BMI 34 is routinely told to lose 15 kilograms before anyone will discuss surgery. Losing weight before a knee replacement is genuinely good advice: it lowers infection and wound-complication risk and improves the outcome. It is not the same thing as being refused care, and the two get conflated. If you are being turned away on BMI alone, ask what the specific risk threshold is, what support is being offered to reach it, and whether the answer would be different at a hospital with a higher-risk surgical service.

When to get a second opinion

Four situations make a second opinion close to mandatory: surgery is recommended at your first visit, before any conservative treatment; you are under 55; your pain is mostly at rest or at night rather than with activity; or your imaging and your symptoms point in different directions. A second opinion costs a consultation fee. A knee replacement you did not need costs a year of recovery, and a knee replacement you needed six months ago costs nothing but time.

16. What a knee replacement really costs in India

The implant inside you is a price-controlled device. The hospital is not. Knowing which part is capped, and by how much, is the difference between a fair bill and an inflated one.

The National Pharmaceutical Pricing Authority has capped the price of orthopaedic knee implants since 2017, when it used extraordinary powers under the Drugs and Prices Control Order to cut implant costs by up to 70%. The ceiling has been extended every year since. As of the most recent extension, in force until 15 November 2026, a primary knee replacement system (titanium alloy coated) is capped at Rs 51,563 and a revision knee replacement system at Rs 83,547.

That figure is the implant, not the operation. It matters because the implant is the line item most often inflated in a private bill, and because you are entitled to ask which implant is being used and what it costs. Ask for the implant sticker. It should be on your discharge summary.

Rs 51,563
Ceiling, primary implant
NPPA cap, in force to Nov 2026

Rs 83,547
Ceiling, revision implant
Same notification

250,000+
Knee replacements a year
India, reported 2025

70%
Implant price cut in 2017
When the ceiling was first imposed

The four cost bands

Hospital charges vary by city, by bed category and by surgeon. The bands below describe what patients commonly report for a single knee, and they are a starting point for a conversation, not a quote. Always get a written estimate with the implant named.

Setting	Typical range for one knee	What drives the difference
Government medical college or district hospital	Free to about Rs 60,000	Subsidised or free implants under schemes; waiting lists; shared rooms; you may buy consumables.
Trust and mid-tier private hospital	Roughly Rs 1.2 to 2.2 lakh	Tier-2 city, standard implant, twin-sharing room, package pricing.
Metro private hospital	Roughly Rs 2 to 3.5 lakh	Tier-1 city, premium implant within the cap, single room, higher surgeon and anaesthesia fees.
Premium or robotic-assisted	Roughly Rs 4.5 lakh and above	Navigation or robotic systems, premium room, international-brand implant, higher facility charges.

Two things are usually not in the quoted package: physiotherapy after discharge, and the treatment of any complication. Ask for both in writing. A year of physiotherapy can add Rs 15,000 to Rs 40,000, and a revision for infection can cost several times the original operation.

Who pays: the schemes most Indian families miss

- **Ayushman Bharat PM-JAY.** Total knee replacement is a listed package. Eligibility is based on the SECC deprivation and occupational categories, and the package rate depends on the hospital's tier. Ask the hospital's Ayushman Mitra for the package code and the rate that applies to you, and confirm the hospital is empanelled for orthopaedics, not just for general medicine.
- **State schemes.** Several states run their own schemes on top of PM-JAY, with different package rates and different empanelled hospitals. The district hospital's medical social worker will know.
- **CGHS, ESIC and railway or defence panels.** If you or a family member is a beneficiary, the procedure is covered at notified rates in listed hospitals. Get the referral and the permission letter before admission, not after.

- **Private health insurance.** Under IRDAI's 2024 health insurance regulations, the waiting period for pre-existing diseases cannot exceed 36 months, and after a 60-month moratorium an insurer cannot contest a claim for non-disclosure except in cases of fraud. If your policy is older than that and the insurer cites a pre-existing condition, ask them to show you the clause.
- **Room rent caps.** A policy that caps room rent at 1% of sum insured usually caps the surgeon's fee and associated charges proportionally. Choose a room within the cap or expect a proportionate deduction.

The pre-admission checklist

1. Written estimate naming the implant, the manufacturer and the price, compared against the NPPA ceiling.
2. Written confirmation of what the package includes and excludes: physiotherapy, blood, consumables, follow-up visits, complications.
3. Pre-authorisation approval from the insurer or scheme, in writing, before you are admitted.
4. A list of your current medicines, including blood thinners, diabetes drugs and supplements, reviewed by the surgeon and the physician.
5. HbA1c, haemoglobin and vitamin D checked and optimised. Poorly controlled diabetes is the single biggest modifiable risk for infection after knee replacement.
6. A dental check if you have untreated dental infection, and treatment of any skin infection, before surgery.
7. A plan for the first two weeks at home: who will help, where you will sleep, how you will use the toilet.

> **The one question that saves the most money**
> "Please write the implant name and its price on the estimate." A hospital that is billing fairly will do it without hesitation. One that will not is telling you where the margin is.

17. The twelve weeks after surgery

The operation is one day. The rehabilitation is twelve weeks, and it decides what you can do for the next twenty years.

What follows is the shape of a typical recovery. Your surgeon's protocol overrides it, and every knee is different. The point is to know what normal looks like, so that you can tell the difference between normal and a problem.

When	What usually happens	What you should be doing
Day 0-1	Standing with a frame or walker, sometimes on the same day. Pain controlled with a combination of medicines.	Ankle pumps every hour, deep breathing, sitting up, standing with help. Ice and elevation.
Day 2-3	Walking short distances with a walker. Wound check. First physiotherapy session.	Straight leg raises, gentle bending, walking to the bathroom. Learn to use the stairs before discharge.
Week 1-2	Swelling peaks and then settles. Staples or sutures out around day 12-14.	Bend to about 90 degrees, straighten fully. Quad sets and heel slides several times a day. Keep the wound dry and clean.
Week 3-6	Walking with a stick or without support, depending on the knee. Stairs become possible.	Stationary cycling with no resistance, sit-to-stand, balance work. Aim for a straight knee and 110 degrees of bend.
Week 6-12	Return to desk work, often between weeks 6 and 8. Driving once you are off strong painkillers and can brake safely.	Progressive strengthening, longer walks, swimming once the wound is fully healed. Most people are off the stick by now.
Month 3-6	Function improves steadily. Some stiffness in the morning is normal.	Full strengthening programme, walking 30 to 45 minutes, stairs without thinking.
Month 6-12	The knee feels like part of you rather than an event.	Maintain strength. Annual review. Report any new pain, swelling or instability.

Setting up the house before you come home

This is the part Indian families under-plan, and it is the difference between a smooth recovery and a fall. Do it in the week before surgery, not the day after discharge.

- **The toilet is the main event.** A commode chair or a raised seat over the Indian pan, plus a grab bar. You will not be able to squat for six to eight weeks, and trying is how people fall.
- **Bed height.** Your feet should touch the floor when you sit on the edge. A low bed means a deep knee bend every time you get up.
- **Clear the floor.** Loose rugs, buckets, wires, stools. A walker needs a clear path from the bed to the bathroom.
- **A chair with arms** in the room where you spend the day. Getting up from a chair without arms is much harder in the first month.
- **Nothing on the floor.** Move the utensils, the pooja items and the stored grain to waist height for three months.
- **Someone at home for two weeks.** Not for company. For the bathroom, the stairs and the first week of exercises.

The complications worth knowing by name

Problem	What it looks like	What to do
Infection	Fever, increasing pain, wound discharge, redness spreading, a knee that gets worse instead of better after week one.	Same-day hospital. Do not wait for the next appointment.
Deep vein thrombosis	Calf swelling, pain or warmth, usually in the first two weeks.	Same-day hospital. This is why you were given blood thinners and stockings.
Pulmonary embolism	Sudden breathlessness, chest pain, coughing blood.	Emergency. Call an ambulance.
Stiffness	Cannot get past 90 degrees of bend by week six.	Tell your surgeon early. Late manipulation is much harder than early physiotherapy.
Persistent pain	Pain that is not improving by three months.	Review with imaging and blood tests. Do not accept "give it time" indefinitely.

Constipation from painkillers is near-universal and rarely mentioned. Fluids, fibre, and a stool softener if needed. Opioid-based pain relief after discharge should be short, and if it is not, that is a conversation to have at the first follow-up rather than a prescription to keep filling.

18. Protecting the knees you have, if you are under 40

Nothing in this chapter is about avoiding activity. It is about not being the person who tore a meniscus at 28 and had a knee replacement at 52.

Recreational running does not appear to increase the rate of knee osteoarthritis in reviews of the evidence, and may be associated with lower rates than a sedentary life. Competitive and elite distance running is a different story, with higher rates reported. The lesson is not "running is bad". It is that dose, surface, footwear, strength and recovery decide the outcome.

The five habits that protect a young knee

1. **Strength before volume.** Before you add kilometres or weight, be able to do 20 sit-to-stands in 30 seconds, 20 single-leg calf raises, and stand on one leg for 30 seconds without wobbling.
2. **Land softly, change direction well.** Most knee injuries that become arthritis are non-contact pivoting injuries in football, basketball and badminton. Landing mechanics and single-leg control are trainable and are the highest-value prevention work available.
3. **Do not ignore a swollen knee.** A knee that swells within 24 hours of a twist has a structural injury until proven otherwise. A meniscal tear at 25 that is never assessed is a common beginning to osteoarthritis at 45.
4. **Replace worn shoes.** A running shoe loses most of its cushioning somewhere between 500 and 700 kilometres. A daily-wear shoe with a worn outer edge changes how you walk.
5. **Break up sitting.** Sedentary work and prolonged sitting were both significantly associated with knee osteoarthritis in the Tiruvallur study. Stand every 45 minutes, take the stairs once a day on purpose.

Gym rules that keep the joint out of trouble

- Squat only as deep as you can keep your heels down and your knees tracking over your toes. For most people with knee pain, that is above parallel, not below it.
- On the leg press, stop before the knee folds past 90 degrees. The last 20 degrees of bend load the kneecap hardest.
- Do not lock the knee hard at the top of a leg extension under heavy load.
- Train the back of the leg and the hip, not only the front. Hamstrings and glutes control the knee.
- If a movement produces pain that lasts more than a day, reduce the range or the load. Do not push through it to "strengthen the joint".

> **The 30-second self-test, once a month**
> Count how many times you can stand from a chair in 30 seconds without using your hands. Below 12 is worth working on. Also time a single-leg stand, and check whether you can walk up two flights without stopping. These three numbers track your knee better than any X-ray.

19. Flares, winter and travel

Knee pain is not steady. It comes in flares, and a flare has a protocol. Knowing it in advance is what stops a bad week from becoming a bad season.

The 48-hour flare plan

1. **Do not go to bed.** Relative rest means fewer stairs and shorter walks, not immobility. Immobility stiffens the joint and weakens the muscle within days.
2. **Cold, not heat, if it is hot and swollen.** A cold pack wrapped in cloth for 15 minutes, three times a day. Heat feels better on a stiff, dry knee; it makes a swollen one worse.
3. **Topical NSAID gel, four times a day.** This is exactly the situation it is for.
4. **Paracetamol or a short course of an oral anti-inflammatory** if you can take one safely, at the lowest dose that works.
5. **Keep the isometrics.** Quad sets and ankle pumps do not irritate a swollen joint and preserve the muscle.
6. **Elevate and compress** if there is visible swelling. A simple crepe bandage, not tight.
7. **Reassess at 72 hours.** If it is not improving, or if there is fever, redness, or you cannot bear weight, see a doctor.

Most flares settle in three to seven days. What turns a flare into a crisis is the story people tell themselves about it: that the knee has deteriorated, that surgery is now inevitable, that they must stop walking for a month. None of that is established by a flare.

Does cold weather make it worse?

Ask any Indian orthopaedic OPD in December and you will hear that it does. The research on weather and osteoarthritis pain is inconsistent, with several studies finding no clear relationship between temperature, pressure and pain. What is well established is that people move less in cold weather, and that a joint which has not been moved is stiffer and more painful. The practical conclusion does not depend on resolving the science: keep exercising indoors, warm up for longer, and dress the knee warmly.

Travel without wrecking the knee

Situation	The problem	What to do
Long flight	Hours with the knee bent at 90 degrees, plus clot risk.	Aisle seat, walk every hour, ankle pumps in the seat, keep the leg straight when you can, water instead of alcohol.
Indian airport	Long walks between terminals and gates.	Wheelchair assistance is free at Indian airports and does not need a certificate. Ask at check-in.
Train	Climbing to an upper berth.	Book a lower berth. It is worth the small premium.
Long drive	Static flexion and a swollen knee at the end.	Stop every 90 minutes, walk 100 metres, adjust the seat so the knee is not fully straight or fully bent.
Wedding season	Floor seating, long standing, dancing, cold floors.	Carry a folding stool, sit at the back where you can stretch, wear cushioned shoes, and take the gel with you.
Pilgrimage	Long walking, stairs, crowds, uneven ground.	Build walking distance for six weeks beforehand, use a stick, plan rest days, and do not attempt the steepest section on day one.

One more travel rule: carry your medicines in your hand baggage, in their original strips, with the prescription. A flare away from home is much worse when your gel is in the checked bag.

20. When to go to hospital today

Almost all knee pain can wait for an appointment. A small number of presentations cannot, and the difference is measured in hours.

Sign	What it may be	How fast
Hot, red, very swollen knee with fever, and you cannot put weight on it	Septic arthritis, an infection inside the joint	Emergency. Same day, emergency department.
A fall, immediate severe pain, inability to stand, deformity	Fracture, particularly in an older adult with osteoporosis	Emergency. Do not try to walk on it.
Knee locked and will not straighten, with sudden severe pain	Displaced meniscal tear or a loose body	Same day or next day.
Calf swelling, pain or warmth, especially after surgery or a long journey	Deep vein thrombosis	Same day.
Sudden breathlessness, chest pain, coughing blood	Pulmonary embolism	Call an ambulance.
New back pain with leg weakness, numbness between the legs, or loss of bladder control	Cauda equina syndrome	Emergency. Hours matter for the outcome.
Night pain that wakes you, with weight loss, fever or night sweats	Infection or, rarely, a tumour	Within days, not weeks.
A knee that becomes hot and swollen over hours, without injury, in a person with gout or kidney disease	Gout or pseudogout	Within a day. Needs diagnosis, not just painkillers.

Where to go matters as much as when. A district hospital or medical college orthopaedic OPD will see you faster and cheaper than a private emergency for anything non-surgical. For advice that is not an emergency, eSanjeevani, the government's free national telemedicine service, connects you to a doctor without a journey. For a second opinion on a surgery recommendation, take your films and your reports to a medical college OPD; the consultation fee is nominal and the incentive to operate is lower.

> **The one-page rule for your fridge**
> Fever plus a hot swollen knee is an emergency. A fall you cannot stand up from is an emergency. A swollen calf is an emergency. Everything else can wait for a weekday appointment, and almost everything else will get better with the programme in this book.

Worksheet 1: Seven-day pain and function diary

Fill this in for one week before your appointment. It converts "it hurts a lot" into something a clinician can act on.

Score pain from 0 (none) to 10 (worst imaginable). Record stiffness as the number of minutes it takes to feel loose after getting up. Record the furthest you walked without stopping, and whether you needed to sit down on the stairs.

Day	Morning pain	After activity	Night pain	Stiffness (min)	Furthest walk	Stairs (easy/hard)	Medicine taken
Mon							
Tue							
Wed							
Thu							
Fri							
Sat							
Sun							

What to look for at the end of the week

- Is the pain worse on a specific day or after a specific activity? That is your load problem.
- Is morning stiffness under or over 30 minutes? Under points to osteoarthritis; over, with swelling, needs a rheumatology opinion.
- Is night pain present? Night pain that wakes you is a reason to see someone sooner.
- Has your walking distance changed over the week? Falling distance with stable pain usually means deconditioning, not damage.

Worksheet 2: The eight-week strength log

Tick each session. Two sessions a week is the minimum that produces change; three is better. Do not add a fourth.

Exercise	W1	W2	W3	W4	W5	W6	W7	W8
Quad set (10 x 5 sec)								
Straight leg raise (10 each)								
Seated knee extension (12 each)								
Heel slide (10)								
Calf raise (15)								
Sit-to-stand (8-12)								
Hamstring curl (12 each)								
Wall sit (3 holds)								
Step-up (10 each)								
Glute bridge (12)								
Side-lying lift (12 each)								
Single-leg balance (3 x 20 sec)								

Monthly progress markers

Test	Start	Week 4	Week 8	Target
Sit-to-stands in 30 seconds				20 or more
Single-leg stand (seconds)				30
Calf raises, one leg				20
Pain-free walking distance (metres)				Double the start
Pain score on stairs (0-10)				3 or less

Worksheet 3: Weight, waist and load tracker

Weigh yourself once a week, same day, same time, after the toilet and before food. Measure your waist at the navel, standing, after breathing out.

Week	Weight (kg)	Waist (cm)	Total lost (kg)	Load off each step (kg)	Notes
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					

Load off each step is your total weight lost multiplied by about four. If you have lost 4 kg, each step carries about 16 kg less. At 5,000 steps a day that is 80,000 kg less load on that knee, every day.

Targets, by Indian cut-offs

If your BMI is	A realistic six-month target	What to change first
23 to 24.9	3 to 4 kg, about 5%	Liquid sugar and evening snacks.
25 to 29.9	5 to 8 kg, about 10%	Portion of grain at two meals, plus a daily walk.
30 or more	10% of body weight	Structured programme with a dietitian or doctor, plus strength work.

Worksheet 4: The doctor visit sheet

Print this page and take it with you. Fill the left column before the visit and write the answers in the right column during it.

My question	The answer I was given
What is my diagnosis, in one sentence?	
Which compartment, and what Kellgren-Lawrence grade?	
What did you find on examination?	
What are the two things you want me to do this month?	
Do I need any scan? What will change if it is positive?	
Which medicine, at what dose, for how many days?	
What should I do on a bad day?	
When should I come back, and what would make you want to see me sooner?	

Bring with you

- All previous X-ray films and reports, not just the newest.
- A list of every medicine you take, including supplements and anything from a chemist without a prescription.
- Your seven-day diary from Worksheet 1.
- Your HbA1c and blood pressure readings if you have diabetes or hypertension.
- One person who lives with you, if possible. Two sets of ears remember more.

Worksheet 5: The surgery decision checklist

Score each line honestly. Six or more clear yes answers means the conversation is reasonable. Fewer means the conservative route has not been tried.

Statement	Yes	No
My pain limits walking, stairs or getting out of a chair most days.		
I have completed at least 12 weeks of structured strengthening.		
I have made a genuine attempt at weight loss if I am overweight.		
I have used a topical NSAID properly for at least two weeks.		
I need painkillers regularly, over months, not days.		
I have rest or night pain, not only activity pain.		
My X-ray shows grade 4 change, confirmed by a standing film.		
I understand the recovery: 12 weeks of rehabilitation, and a year to full benefit.		
I have a written estimate naming the implant and the all-in cost.		
I have had a second opinion, or I am under 55 and have deliberately decided not to.		

If you answer no to any of the first five

Ask for those things before you book a date. The ICMR's own indication is surgery after conservative treatment has failed, and a knee replacement is a good operation that is made worse by being done too early, in a patient who has not had the chance to avoid it.

Worksheet 6: The annual knee spend audit

Most households cannot say what they spent on their knee last year. This page fixes that in ten minutes.

What I bought	Cost (Rs)	Did it help?	Keep or stop

Where the money should go, in order

1. A commode stool or raised seat, and a grab bar. Under Rs 3,000, and it removes the deepest squat of your day.
2. A resistance band and cushioned walking shoes.
3. Six to eight supervised physiotherapy sessions, then a home programme.
4. A topical NSAID gel, used consistently for two weeks.
5. A walking stick, if balance or distance is a problem.
6. A gym or pool membership, if you will actually use it.
7. Then, and only then, anything in a bottle with a health claim on it.

Worksheet 7: The flare card and the red-flag card

Cut this page out, or photograph it and keep it on your phone. It is for the day you cannot think clearly because it hurts.

My flare plan, first 48 hours

1. Fewer stairs, shorter walks, no bed rest. 2. Cold pack, 15 minutes, three times a day, wrapped in cloth. 3. Topical gel four times a day. 4. Paracetamol or my short-course anti-inflammatory, at the dose my doctor approved. 5. Quad sets and ankle pumps, twice a day. 6. Elevate if swollen. 7. Reassess at 72 hours.

Go to hospital today if

The knee is hot, red and swollen with fever, and I cannot put weight on it. I fell and cannot stand. My calf is swollen and painful. I am suddenly breathless or have chest pain. My knee is locked and will not straighten. I have new leg weakness or numbness with back pain.

My numbers, filled in

My diagnosis	
My grade and compartment	
My doctor and hospital	
My physiotherapist	
Medicines I can take safely	
Medicines I must avoid	
My current weight and waist	
My target weight	
Emergency contact	

Sources and further reading

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Market, pricing and regulatory reporting

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- Business Standard. NPPA extends knee implant price ceiling by one year till November 2026, 17 November 2025. Includes the ceiling figures and the 250,000-plus annual knee replacement estimate.
- Journal of Law and the Biosciences, 2025. Indian regulatory framework and the surge of unproven stem cell therapies.
- Times of India, March 2025. Telangana Medical Council review of PRP therapy allegations following complaints from 75 orthopaedic surgeons.

A note on what changes

Guidelines are revised. Prices change with every notification. Drug availability varies by state. Before acting on any number in this book, check the current version: the ICMR STW page, the NPPA medical devices page, and your own insurer's policy wording. The principles in this book have been stable for a decade. The numbers have not.